02 Jun 2026 11:16:54 ET | 13 pages

US Semiconductors and Hardware

Silicon Valley Bus Tour Takeaways

CITI'S TAKE

We hosted investor meetings with AMD, AMAT, ALAB, NVMI, STX, WDC, and SMCI as part of Citi's annual Silicon Valley bus tour yesterday. Please see detailed notes inside.

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Silicon Valley Bus Tour Notes

AMD

Jean Hu, CFO & Treasurer | Matt Ramsay, CVP Financial Strategy & IR | Prabh Gowrisankaran, Director, IR

- **Server growth.** Inflection of Agentic AI and rise of CPU demand was noticeable last year. AMD articulated the 3 segments of server CPUs at investor day last November and was preparing for a significant ramp. Now AMD is seeing an explosion of demand and is working to expand supply. AMD grew server revenue >50% in Q1. TSMC 3nm has long lead times, but they feel good about 2026 and 2027. They are seeing enterprises adopting agentic AI. AMD is seeing the industry shift towards inference and the shift from chatbot inference towards agentic inference. AMD believes their agentic share could exceed their general server CPU share.

- **ARM vs X86 debate.** AMD wants to build the best server platform for different applications and workloads. Turin has a breadth of 8 cores to 192 cores to support different workloads. General purpose CPU, AI head nodes, and Agentic all have different requirements. Agentic AI is multiple workloads, orchestration, data base retrieval, agent management and more. AMD's investments for more than a decade allows them to address all these workloads. AMD is in every workload for servers besides NVL head nodes. Venice CPU is the head node for Helios. AMD believes Venice will be a leadership performance product and Venice will have Verano to compete with ARM CPUs. Verano was closely designed with Meta. X86 security features are strong compared to ARMs.

- **Pricing power.** AMD wants to be strategic to manage long-term relationships with customers in the future. They will share memory cost increases with customers. They have good gross margins in servers. 70% to 75% of AMD server growth would predict to be from units. ASP expansion will come from a greater mix of Turin, and higher core count products. Mandate from AMD management isn't on pushing on X86, it's about building the best CPU and having SKUs to target the whole market.

- **Software.** Largest AI frontier companies are using NVDA, AMD, and different ASICs. AMD is seeing Claude code internally to shorten software deployments. AMD believes the CUDA moat will get diminished given the velocity of software. On the CPU side, its more about the architecture of the platform.

- **Agentic adoption spiked in February.** Model capability increased, and agentic adoption increased when they saw the capability increase. By January of this year, all major hyperscale companies significantly increased their demand on the CPU side. Agentic is happening much faster than AMD expected. Orchestration and execution are all driven by CPUs. 95% of AMD token spending is for software engineering. If AMD believes their customers bottom-up forecast, their $120bn TAM estimate is too low.

- **Intel competition.** Diamond Rapids doesn't have multithreading and ARM AGI does not have multithreading. AMD's enterprise share shift accelerated rapidly in 1Q and one thesis is because customers know Intel doesn't have 8 channel.

- **Networking.** AMD feels good where they are at as they will rely on partners for scale across. Infinity Fabric will be used for scale up. AMD donated UAL to the industry body. Astera and Marvell will be building ethernet switching technology with AMD. Copper and Optics will coexist for a few generations.

- **AI-driven PC Refresh.** AMD isn’t seeing signs yet, but there are different types of tokens that need to be optimized. You can in theory lower your enterprise token budget by running them on a notebook.

AMAT

Prabu Raja, PhD, President, Semiconductor Products Group

- **DRAM/HBM growth.** AMAT continues to see strong demand for DRAM. They did see some absorption of HBM last year and now its opening up. Systems will grow >30% this year, NAND/DRAM leading edge will be growing faster than that.

- **Cyclicality.** PC/Phones people bought every two years in a cycle. Now demand is not following a cycle. The number of steps are growing for logic and DRAM.

- **Industry dynamics.** AMAT has been talking about inflections for some time. Logic is going to GAA and Backside Power. CPUs to GPUs connects a lot more transistors and connects a lot more wiring layers. DRAM is really following the Logic roadmap. Chips are a lot more complex. Wiring has CVD, PVD, and CMP. CPU chip goes from 17 layers to 21 layers for GPU.

- **Advanced packaging.** AMAT started investing more than 10 years back. Bonding is the next inflection, and wafer level panel. AMAT is not strong in plating and that is an opportunity for AMAT. Moreover, display business is synergistic with panels.

- **Terafab.** AMAT can't talk about specific customers. There is broadening of demand that is good for the industry. AMAT has doubled its manufacturing footprint, Singapore has a lot of space to fill tools. Scaling AMAT’s manufacturing is very doable in this environment.

- **Lead times.** The biggest focus is supply chain readiness. The other area of focus is tool startup, is AMAT hiring people early enough and are they trained well enough. AMAT can increase their manufacturing capacity easily right now.

- **ICAPs**. They are seeing this market improve. ICAPs should grow mid to high single digits long term, but they see more growth in direct AI data center.

- **NAND capacity increase timing.** Customers are spending on NAND and NAND will grow this year off a lower base. Demand is good, customers do buy technology upgrades today. NAND fabs were converted to DRAM. AMAT doesn’t see new NAND fabs coming up right now, but they believe it will happen.

ALAB

Des Lynch, CFO | Mike Tate, Strategic Advisor

- **ALAB primer.** The company solves connectivity. They cited a 2030 opportunity of $25 billion. The largest opportunity is switching and their Scorpio P-Series (scale out) and X-Series (scale up) which is a 10 billion opportunity. They expect Scorpio will be largest product family by end of the year. One of key products was the retimer family under Gen 5, now they are transitioning to gen 6, typically moving from one generation gives you 25-30% ASP benefit. AEC is 10-15% of sales, and the company has grown their 200G and 400G solutions, and they see a transition to 800G occurring later this year. CXL is also having growing interest with KV Kache, more of a 2027/2028 impact for the model.

- **NVDA PCIE Gen 6.** Moving to Gen 6 is favorable for ALAB as they are shipping in volume for this technology. Nvidia uses a proprietary scale up technology, and they use Scorpio for scale out.

- **CXL.** $4 billion opportunity by 2030, the company is encouraged by recent engagements, but AI has pushed some of this right. CXL is used by general purpose servers, it enables an expansion of memory.

- **Scorpio P and X engagements.** Scorpio P is driven by ALAB's lead customer, they expect to see two hyperscalers convert to revenue later this year and into next year. Scorpio X has 10 engagements across hyperscalers and AI platform providers. They expect to convert some of these into design wins and revenue.

- **Content.** Aries less than $100 per XPU, Aries plus a few hundred dollars, w/Scorpio P $500, and w/Scorpio X over $1000 per XPU. Dollar content continues to grow and content varies on customers: Google has a proprietary technology and NVDA has their own.

- **Photonics.** Rack to rack connectivity will be first to use optics. 2027 will use NPO and NPO will be dominate in 2028. May see CPO in 2028 depending on how it works. Next generation of XPUs will be optical, CPO won't be primetime yet. People want to stay on copper as long as possible.

- **ALAB growth drivers.** Customer position will drive a lot of growth. ALAB is engaged with all the hyperscalers, and merchant GPU providers like AMD/NVDA. Volumes are more material when you do scale out and scale up. They see continued growth beyond just Amazon.

- **$10 billion Scale Up.** This was a bottom-up exercise to come up with the $10 billion estimate. Amazon and AMD are the two main endorsers of UAL.

- **Market Share.** Historically had 90% share in retimers, only loss is from China. In Gen 6, AVGO announced theirs, but ALAB hasn't seen them shipping or MRVL. 1/3rd of ALAB revenue in Q1 was retimers and they feel like they have a good lead. On switching, this will be a competitive environment. On AEC, Credo has larger market share today. MRVL is also making a play in AEC ethernet as well.

- **M&A.** ALAB will continue to look to enhance their portfolio, they did some smaller acquisitions and one at the end of last year. They look to do these acqui-hires as well.

- **Automotive opportunity.** Auto has been very tempting for ALAB, there is a lot of opportunity in PCIE retimers, but now they are focused on AI. ALAB was close to entering this market a few times.

- **Target Model.** They want to grow faster than the market. They talked about 70% gross margins, Q1 was 76%. Some products are accretive and some are dilutive to this margin profile. They target 40% operating income. By growing faster than the market they mean growing faster than hyperscaler capex growth.

- **Equity investments.** Amazon has been doing this since 2014 and only with strategic partners. They signed up for $6.5 billion which shows the revenue opportunity for ALAB. Given ALAB's balance sheet they aren't looking for investments right now.

- **TSMC.** They use TSMC in their front end. On back end they have diversified supply chains.

- **Cosmos Software.** This is very differentiated software and used for two main use cases: design and fleet management software.

NVMI

Gaby Waisman, President & CEO | Guy Kizner, CFO | Adrian Wilson, President & General Manager, Material Metrology Division

- **Key Metrics.** Revenue last year was $881M, up 31% YoY with EPS of $8.62, up 29% YoY. The company has over 7300 active systems with over 400 customer sites. From 2020-2025 CAGR, Nova outperformed WFE growth by 96%.
- **TCO.** TCO is mostly driven by increased throughput. If a Nova tool has a 20% higher price, it needs to come with a 20-25% throughput to win in the market.
- **Q1 Results.** Nova had record revenues in Q1 driven by memory device demand for HBM and DRAM and expects strong growth in 2H26 and 2027 as well. Nova is being more conservative versus other peers because of visibility. Nova's goal is to outgrow WFE.
- **Growth drivers.** GAA is a tailwind, and backside power is one. 3D DRAM, 3D NAND, Hybrid bonding and advanced packaging are all positives for Nova. The more complicated the process, the better it is for a company like Nova.
- **Challenges**. Logic, Memory, and Packaging all have dimension, material, and chemical challenges. Nova will invest now to address the challenges down the road.
- **Strategy.** The company invests more than 15% of sales into R&D to maintain tech leadership, fund new growth engines, and path finding and incubation.
- **Target model.** Operating margin of 28-33% and EPS of $10. Revenue of >$1 billion with gross margins of 57-60%. The company will release a new target model when they reach their current targets.
- **Growth rates.** Advanced packaging is growing faster than front end. Nova has 100% market share in advanced packaging metrology today.
- **Hybrid bonding opportunity**. It is introducing more complexity, and the company sees market share gains moving from one architecture to another in advanced packaging and in hybrid bonding. Hybrid bonding is an inflection point and an opportunity to improve their competitive position.
- **Services**. 21% of Nova revenue is in services, within the range of 20-25%. This will fluctuate alongside with tool purchases.
- **Memory price pressure.** Memory prices are going up and is a challenge. The company will work with customers, but they see pricing going up dramatically.

SMCI

Michael Staiger, SVP of Corporate Development | Krishna Shankar, VP Finance | Ian Tolle, Sr. Manager, IR

- Agentic AI is seeing broad adoption and underpinning demand momentum across enterprises/sovereigns/neoclouds.

- A variety of chipset offerings that can be customized to suit client needs and delivering a full stack of datacenter building block solutions (server, storage, switching with software and services attach) is a meaningful driver for SMCI's profitability.
- DCBBS is expected to be 15-20% of profits from low single digits and a higher mix of DCBBS should also help reduce revenue volatility.
- Customer diversification will occur over time though near term can be impacted by some large orders from larger customers that need to be fulfilled.
- Demand from neoclouds has diversified beyond a handful of customers to several others internationally.
- Enterprises are much more focused on getting access to systems (vs focused on price) especially in certain industries like financial industries, oil and gas
- SMCI's vertical integration enables us to alleviate some of the supply constraints and meet customer demand .
- Focus is on optimizing working capital so capital raise is not a requirement to meet growth expectations.

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